AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX

ANNUAL EXAMINATIONS (THEORY) 2025

General Mathematics Paper II

Time: 1 hour 40 minutes Marks: 30

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are NINE questions. Answer ALL questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.

6. You may use a simple calculator if you wish.

List of Formulae

Note:

The symbols have their usual meanings.

Business Mathematics

$$\text{Loss }\% = \left(\frac{\text{Loss}}{CP} \times 100\right) \qquad \text{Profit }\% = \left(\frac{\text{Profit}}{CP} \times 100\right) \qquad SP = CP \times \left(\frac{100 + \text{profit}\,\%}{100}\right)$$

$$SP = CP \times \left(\frac{100 - \text{loss}\%}{100}\right) \qquad \text{Discount }\% = \frac{\text{Discount}}{MP} \times 100$$

Sets and Functions

$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

Exponents and Logarithms

$$a^m \times a^n = a^{m+n} \qquad a^m \div a^n = a^{m-n} \qquad \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$

$$\left(a^m\right)^n = a^{mn} \qquad (a \times b)^m = a^m \times b^m \qquad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$\log_a (m)^n = n \log_a m \qquad \log_a (m \times n) = \log_a m + \log_a n \qquad \log_a \left(\frac{m}{n}\right) = \log_a m - \log_a n$$

$$\log_a n = \log_b n \times \log_a b \qquad \log_a b = n \Leftrightarrow a^n = b$$

Algebraic Formulae and Applications/ Factorisation

$$(a-b)^2 = a^2 - 2ab + b^2 \qquad (a+b)^2 = a^2 + 2ab + b^2$$

$$a^2 - b^2 = (a+b)(a-b) \qquad 2\left(a^2 + b^2\right) = (a+b)^2 + (a-b)^2$$

$$(a+b)^2 - (a-b)^2 = 4ab \qquad (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

Linear Graphs

$$1 \text{ mile} = \frac{8}{5}\text{km} \qquad 1 \text{ Hectare} = 2.471 \text{ Acres} \qquad {}^\circ F = \left(\frac{9}{5} \times {}^\circ\text{C}\right) + 32$$

Matrices and Determinants

$$A^{-1} = \frac{1}{|A|} AdjA$$

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(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 3 Marks)

a. AB Studio offers the following discounts as tabulated below.

Quantity	Price Before Discount (Rs)	Discount Percentage
First 10 Prints	150	10%
Next 20 Prints	260	15%

If a client requests for 30 prints, then calculate the following:

i. total discount offered. (2 Marks)

ii. amount to be paid after discount. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.1.)

b. The property inherited by Jamal was distributed among his widow, two daughters and a son.

If widow received Rs 300,000, then calculate the

(**Note**: The share of a son is two times that of a daughter.)

i. total worth of the property. (1 Mark)

ii. share of each daughter. (2 Marks)

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Q.2. (Total 4 Marks)

A fruit vendor bought 600 apples for Rs 4,800. He spent Rs 500 on transportation. Find the price at which he should sell each apple to get a total profit of Rs 1,900. Also, find his profit percentage.

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Q.3. (Total 3 Marks)

If two sets are $A = \{1, 2\}$ and $B = \{b, c\}$, then

i. find $A \times B$. (1 Mark)

ii. write one function from *A* to *B*. (1 Mark)

iii. write one relation from A to B that is not a function. (1 Mark)

Q.4. (Total 3 Marks)

Applying laws of logarithms, expand the expression $\log \sqrt[3]{\frac{(a+b)^2}{c}}$.

Q.5. (Total 4 Marks)

For the expressions $2x+y+1=4$ and $2xy-y+2x=6$, find the value of $4x^2+y^2+1$.

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 4 Marks)

a. Prove that the factors of $x^4 + 64$ are $(x^2 + 4x + 8)$ and $(x^2 - 4x + 8)$.

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ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 4 Marks)

b. For the polynomial $x^3 - 2x^2 - 5x + 6$,

i. show that 1 and -2 are zeros of the given polynomial. (2 Mark)

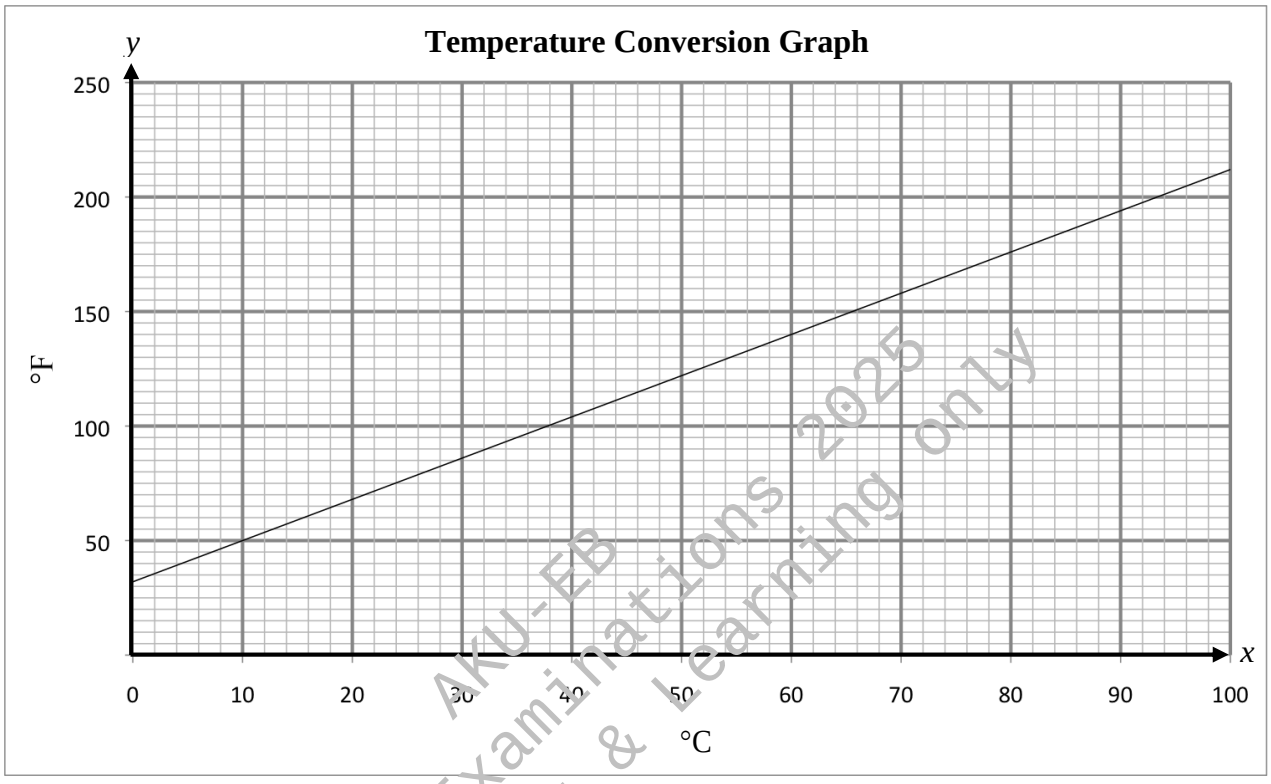
ii. find the remaining third zeros of the polynomial. (1 Mark)

iii. write the factors in a factorised form. (1 Mark)

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Q.7. (Total 3 Marks)

The given graph shows relationship between the temperature measured in degree °F and degree °C scales.



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Using the information in the graph, convert

i. 0° C into °F. (1 Mark)

ii. 10° C into °F. (1 Mark)

iii. 100°F into °C. (1 Mark)

(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 3 Marks)

a. If $A=\begin{bmatrix}1 & -1\\ 0 & 2\end{bmatrix}$ and $B=\begin{bmatrix}2 & 1\\ 0 & 1\end{bmatrix}$, then prove that $(AB)^t = B^t A^t$.

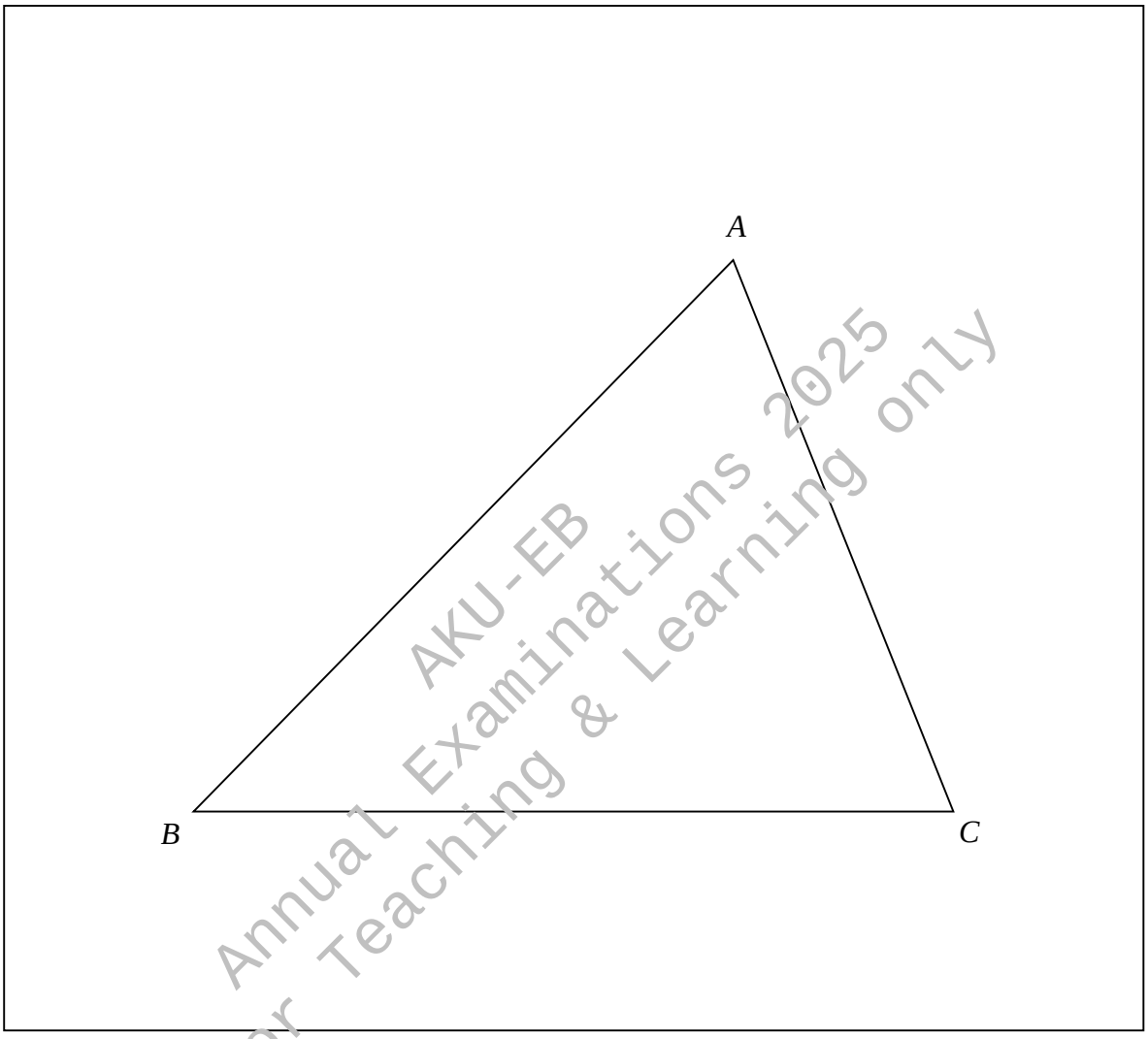
b. Solve the given matrix equation to find the unknown matrix X.

$$2X-\begin{bmatrix}1 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}2\\ 3\end{bmatrix}=\begin{bmatrix}-1\\ 1\end{bmatrix}$$

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Q.9. (Total 3 Marks)

The given diagram shows a triangle *ABC*.



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Show that the angle bisectors of the given triangle are concurrent.

Also, mention the point of concurrency.

END OF PAPER

Please use this page for rough work

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